

EE-SE Pareto Boundary and Amplification Structure Optimization for Hybrid STAR-RIS

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Abstract—Hybrid active-passive simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) can improve spectral efficiency (SE) relative to fully passive architectures, but their energy efficiency (EE) behavior remains insufficiently understood because existing analyses often rely on simplified amplifier-noise models, simplified control and switching power models, and asymmetric task models for the transmission and reflection sides. This paper develops a structured analytical framework for the EE-SE Pareto boundary of hybrid active-passive STAR-RISs. We first introduce a three-part power model that accounts for static controller and bias power, dynamic switching overhead, and communication-side transmit and amplification power. For a sparse active set, we derive a closed-form SE-optimal amplification structure and an active-on threshold that quantifies when amplification benefits outweigh amplifier noise. We then characterize the Pareto boundary via an ϵ -constraint formulation with a low-complexity sparse-activation solver and show that the resulting coefficient rules extend across ES, TS, and SS modes. Numerical results under the simulated settings show that the proposed sparse-hybrid design improves EE over the considered baselines.

Index Terms—STAR-RIS, active RIS, hybrid active-passive RIS, energy efficiency, spectral efficiency, Pareto boundary, amplifier noise.

I. INTRODUCTION

Simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) further extend passive beam manipulation from one-sided reflection to 360° coverage, enabling distinct user groups on the transmission and reflection sides to be served by a common surface [1]–[3]. This dual-side coverage benefit is especially attractive when transmission-side and reflection-side users have different path losses, quality-of-service (QoS) targets, and scheduling roles. Therefore, as shown in Fig. 1, the hybrid STAR-RIS can simultaneously provide transmission communication for indoor users and reflection communication for outdoor users. It is thus an ideal solution for indoor-outdoor hybrid coverage.

A persistent limitation, however, is the multiplicative path loss of purely passive surfaces. Active and hybrid active-passive RIS architectures alleviate this loss by embedding a small number of amplifying elements, but they also inject amplifier noise and incur non-negligible bias power [4]–[11]. Recent measurement-driven studies further show that RIS power consumption is not a single constant; it contains static controller power, technology-dependent per-element consumption, and dynamic state-switching overhead [12]–[14]. These observations become even more consequential for STAR-RIS because transmission and reflection generally serve different

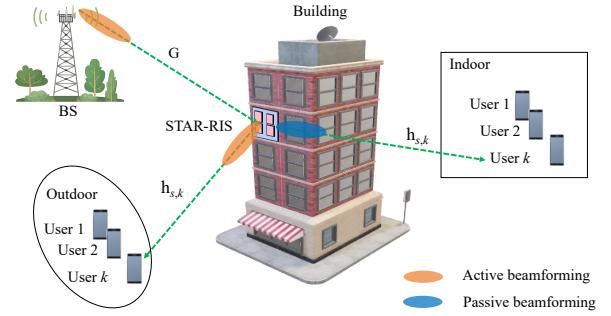


Fig. 1. The hybrid STAR-RIS-aided wireless system.

tasks and thus compete for the surface aperture, side-wise power split, and active hardware budget.

Despite rapid progress, three gaps remain open. *First*, the noise injected by active elements is often modeled coarsely, so there is still no transferable rule explaining when active mode should be used, how many active elements are needed, and whether they should preferentially support the transmission or reflection side. *Second*, EE conclusions are frequently based on oversimplified power models, which makes them difficult to map to the control and hardware layers. *Third*, existing hybrid active-passive STAR-RIS studies are still dominated by numerical optimization, while structural Pareto-boundary characterizations and unified coefficient rules across energy splitting (ES), time switching (TS), and space splitting (SS) are scarce [15]–[20].

To this end, we develop an equivalent formulation for the hybrid STAR-RIS model and use it to derive tractable Pareto-boundary characterizations. The main contributions are as follows.

- We formulate a hybrid active-passive STAR-RIS downlink with users on the transmission and reflection sides, and we introduce a three-part power model including static controller and bias power, dynamic switching power, and communication-side BS transmit and amplifier power.
- For a fixed active set, we derive an equivalent side-wise SINR model. The resulting SE-optimal common amplification factor admits a closed-form expression, and the derivative yields an *active-on threshold* that quantifies when the coherent-gain benefit dominates amplifier-noise injection.
- We develop a structured sparse-activation solver whose

outer selection, side-aware weighting, closed-form gain update, and target-rate inversion remain analytically consistent with the proposed framework. This yields a low-complexity alternative to mixed-integer nonlinear programming.

- Our numerical study indicates the following design implications: passive STAR-RIS is competitive at low SE, sparse hybrid activation is preferable at moderate-to-high SE, the optimal active ratio under ES grows from about 0.13 at 8 bit/s/Hz to about 0.60 at 20 bit/s/Hz, and stronger amplifier noise lowers the preferred gains most visibly under TS while narrowing the EE region.

II. SYSTEM MODEL

A. Hybrid active-passive STAR-RIS downlink

Consider a BS with M antennas serving K_t users on the transmission side and K_r users on the reflection side via an N -element STAR-RIS. Let $\mathcal{K}_t = \{1, \dots, K_t\}$ and $\mathcal{K}_r = \{1, \dots, K_r\}$. Under the ES protocol, the n th STAR-RIS element uses transmission and reflection coefficients

$$\theta_{n,s} = \sqrt{\rho_{n,s}} \eta_{n,s} e^{j\phi_{n,s}}, \quad s \in \{t, r\}, \quad (1)$$

with $\rho_{n,t} + \rho_{n,r} = 1$, $0 \leq \rho_{n,s} \leq 1$, and $1 \leq \eta_{n,s} \leq \beta_{\max}$. For passive elements, $\eta_{n,s} = 1$; for active elements, $\eta_{n,s} = \beta_{n,s} > 1$ on the assigned side. Let $\mathcal{A}_t$ and $\mathcal{A}_r$ denote the active-element sets for transmission and reflection, and let $\mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r$.

Denote by $\mathbf{G} \in \mathbb{C}^{N \times M}$ the BS-to-STAR-RIS channel, and by $\mathbf{h}_{s,k} \in \mathbb{C}^N$ the STAR-RIS-to-user channel on side s for user k . The received signal of user (s, k) is

$$y_{s,k} = \mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \sum_{j \in \mathcal{K}_s} \mathbf{w}_{s,j} x_{s,j} + \mathbf{h}_{s,k}^H \mathbf{n}_{a,s} + n_{s,k}, \quad (2)$$

where $\Theta_s = \text{diag}(\theta_{1,s}, \dots, \theta_{N,s})$, $\mathbf{w}_{s,j}$ is the BS beamformer for user (s, j) , $x_{s,j}$ is the unit-power information symbol, $n_{s,k} \sim \mathcal{CN}(0, \sigma_0^2)$ is thermal noise, and $\mathbf{n}_{a,s}$ collects the amplifier noises of the active elements on side s . We model

$$\mathbf{n}_{a,s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{a,s}), \quad \mathbf{C}_{a,s} = \text{diag}(c_{1,s}, \dots, c_{N,s}), \quad (3)$$

where $c_{n,s} = \rho_{n,s} \beta_{n,s}^2 \sigma_a^2$ for $n \in \mathcal{A}_s$ and $c_{n,s} = 0$ otherwise. The SINR of user (s, k) becomes

$$\gamma_{s,k} = \frac{|\mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \mathbf{w}_{s,k}|^2}{\sum_{j \neq k} |\mathbf{h}_{s,k}^H \Theta_s \mathbf{G} \mathbf{w}_{s,j}|^2 + \sigma_{a,s,k}^2 + \sigma_0^2}, \quad (4)$$

where $\sigma_{a,s,k}^2 = \mathbf{h}_{s,k}^H \mathbf{C}_{a,s} \mathbf{h}_{s,k}$ is the active-noise term seen at user (s, k) .

B. Three-part power-consumption model

Motivated by recent measurement-based RIS power studies [12], [13], we write the total consumed power as

$$\begin{aligned} P_{\text{tot}} = & \frac{1}{\eta_{\text{PA}}} \sum_{s \in \{t, r\}} \sum_{k \in \mathcal{K}_s} \|\mathbf{w}_{s,k}\|^2 + P_{\text{ctrl}} + P_{\text{bias}} |\mathcal{A}| \\ & + f_u(P_{\text{sw},0} + P_{\text{sw},1} |\mathcal{A}|) \\ & + \sum_{n \in \mathcal{A}_t} \mu_t(\beta_{n,t}^2 - 1) + \sum_{n \in \mathcal{A}_r} \mu_r(\beta_{n,r}^2 - 1), \end{aligned} \quad (5)$$

where the first term is the BS radiated power divided by PA efficiency η_{PA} , P_{ctrl} is the controller power, P_{bias} captures static per-active-element biasing, f_u is the state-update frequency, and the last two sums describe the amplification-dependent power. This decomposition explicitly distinguishes static, dynamic, and communication-side costs and is later used to interpret the representative power breakdown in Fig. 5 and the hardware-sensitivity trends in Fig. 4.

C. Structured restricted equivalent formulation

To obtain interpretable structure, we adopt an restricted equivalent model in which the BS uses side-wise zero-forcing directions and only the scalar user powers remain to be allocated. The STAR-RIS phases are aligned to the dominant cascaded paths, and all active elements on the same side share a common gain β_s . Let g_n be the effective BS-to-element channel after zero-forcing projection, and let $h_{s,k,n}$ be the n th entry of $\mathbf{h}_{s,k}$. Define the equivalent coherent-gain quantity

$$u_{s,k,n} = |g_n| |h_{s,k,n}|. \quad (6)$$

Then the side-wise effective coherent term for user (s, k) is

$$\psi_{s,k}(\beta_s) = \sum_{n \in \mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n} + \beta_s \sum_{n \in \mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \quad (7)$$

where $\mathcal{P}_s = \{1, \dots, N\} \setminus \mathcal{A}_s$ includes passive elements and the elements that are active for the opposite side but only passive for side s . The corresponding active-noise term is

$$\zeta_{s,k}(\beta_s) = \sigma_0^2 + \rho_s \beta_s^2 \sigma_a^2 \sum_{n \in \mathcal{A}_s} |h_{s,k,n}|^2. \quad (8)$$

To obtain interpretable structure, we further specialize the side targets to equal per-user rate splitting within each side. Therefore, the closed-form developments below capture side-wise asymmetric traffic demands, but not arbitrary heterogeneous user-specific QoS within each side. The minimum sum transmit power needed to satisfy a side target rate R_s under equal user-rate splitting is then

$$p_s^{\min}(\beta_s) = \sum_{k \in \mathcal{K}_s} \frac{(2^{R_s/K_s} - 1) \zeta_{s,k}(\beta_s)}{\psi_{s,k}^2(\beta_s)}. \quad (9)$$

The numerical implementation uses (9) for power inversion together with the structural gain rule derived next.

III. PROBLEM FORMULATION

The sum spectral efficiency is

$$R_{\Sigma} = \sum_{s \in \{t, r\}} \sum_{k \in \mathcal{K}_s} \log_2(1 + \gamma_{s,k}), \quad (10)$$

and the system energy efficiency is

$$\text{EE} = \frac{R_{\Sigma}}{P_{\text{tot}}} \quad (\text{bit/J}). \quad (11)$$

The original EE-maximization problem can be written as

$$\begin{aligned}
& \max_{\mathbf{w}, \rho, \phi, \beta, \mathcal{A}} \quad \frac{R_\Sigma}{P_{\text{tot}}} \\
& \text{s.t.} \quad R_{s,k} \geq \bar{R}_{s,k}, \quad \forall s, k, \\
& \quad \sum_{s,k} \|\mathbf{w}_{s,k}\|^2 \leq P_{\text{max}}^{\text{BS}}, \\
& \quad 1 \leq \beta_{n,s} \leq \beta_{\text{max}}, \quad \forall n \in \mathcal{A}_s, \\
& \quad \beta_{n,s} = 1, \quad \forall n \notin \mathcal{A}_s, \\
& \quad \rho_{n,t} + \rho_{n,r} = 1, \quad 0 \leq \rho_{n,s} \leq 1, \\
& \quad \rho_{n,s} \beta_{n,s}^2 \sigma_a^2 \leq \Gamma_{n,s}, \quad \forall n, s,
\end{aligned} \tag{12}$$

where $\Gamma_{n,s}$ is a maximum noise-injection or hardware-safety limit. Problem (12) is highly non-convex because of the fractional objective, binary active-set selection, and the multiplicative coupling between beamforming, STAR coefficients, and active noise.

To trace the EE-SE Pareto boundary, we adopt the ϵ -constraint formulation

$$\begin{aligned}
\mathcal{P}(\epsilon) : \quad & \min_{\mathbf{w}, \rho, \phi, \beta, \mathcal{A}} \quad P_{\text{tot}} \\
& \text{s.t.} \quad R_\Sigma \geq \epsilon, \\
& \quad \text{constraints of (12)}.
\end{aligned} \tag{13}$$

Every feasible solution of (13) yields one point on the Pareto boundary. The remaining question is how to solve (13) with enough structure to produce transferable design rules.

IV. OPTIMAL STRUCTURE OF THE AMPLIFICATION FACTORS

A. Equivalent side-wise SINR law

For a fixed active set, define

$$a_s = \min_k \sum_{n \in \mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n}, \quad b_s = \min_k \sum_{n \in \mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \tag{14}$$

and

$$c_s = \sigma_0^2, \quad d_s = \max_k \rho_s \sigma_a^2 \sum_{n \in \mathcal{A}_s} |h_{s,k,n}|^2. \tag{15}$$

With average side power $\bar{p}_s$, a conservative worst-user SINR lower bound is

$$\gamma_s(\beta_s) = \frac{\bar{p}_s(a_s + b_s \beta_s)^2}{c_s + d_s \beta_s^2}. \tag{16}$$

Equation (16) isolates the key conflict: coherent signal growth is linear in β_s , while injected active noise grows quadratically.

Proposition 1. For fixed (a_s, b_s, c_s, d_s) with $a_s > 0$, $c_s > 0$, and $d_s > 0$, the function $\gamma_s(\beta_s)$ in (16) is unimodal over $[1, \beta_{\text{max}}]$, and its SE-optimal gain is

$$\beta_{s,\text{SE}}^* = \left[\frac{b_s c_s}{a_s d_s} \right]_{[1, \beta_{\text{max}}]}, \tag{17}$$

where $[x]_{[l,u]} = \min\{u, \max\{l, x\}\}$.

Proof: Differentiating (16) gives

$$\frac{d\gamma_s}{d\beta_s} = \frac{2\bar{p}_s(a_s + b_s \beta_s)(b_s c_s - a_s d_s \beta_s)}{(c_s + d_s \beta_s^2)^2}. \tag{18}$$

The derivative changes sign only once, at $\beta_s = b_s c_s / (a_s d_s)$, which proves unimodality and (17). ■

A direct corollary of (18) is the *active-on threshold*:

$$\left. \frac{d\gamma_s}{d\beta_s} \right|_{\beta_s=1} > 0 \iff b_s c_s > a_s d_s \iff \frac{b_s}{a_s} > \frac{d_s}{c_s}. \tag{19}$$

Equation (19) answers *when* active mode should be used: amplification is worthwhile only if the relative coherent gain brought by the candidate active set exceeds the normalized amplifier-noise penalty. At high SNR (small effective c_s), the threshold becomes harder to satisfy; hence aggressive amplification is less attractive because active noise dominates.

B. Power-minimizing gain and a sufficient convexity condition

For a target side rate R_s , (16) implies the lower-bound power cost

$$\Psi_s(\beta_s) = \frac{K_s(2^{R_s/K_s} - 1)(c_s + d_s \beta_s^2)}{\eta_{\text{PA}}(a_s + b_s \beta_s)^2} + \mu_s |\mathcal{A}_s| (\beta_s^2 - 1). \tag{20}$$

Its first derivative is

$$\Psi'_s(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)(a_s d_s \beta_s - b_s c_s)}{\eta_{\text{PA}}(a_s + b_s \beta_s)^3} + 2\mu_s |\mathcal{A}_s| \beta_s, \tag{21}$$

and its second derivative is

$$\Psi''_s(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)}{\eta_{\text{PA}}(a_s + b_s \beta_s)^4} (a_s^2 d_s - 2a_s b_s d_s \beta_s + 3b_s^2 c_s) + 2\mu_s |\mathcal{A}_s|. \tag{22}$$

Boundary cases: The sufficient condition below is only needed for the mixed case $a_s > 0$ and $b_s > 0$. The degenerate cases are handled separately. If $|\mathcal{A}_s| = 0$, then $b_s = d_s = 0$ and side s is fully passive, so there is no active gain to optimize and we set $\beta_s = 1$ by convention. If $b_s = 0$ while $a_s > 0$ and $|\mathcal{A}_s| > 0$, then

$$\Psi'_s(\beta_s) = 2\beta_s \left(\frac{K_s(2^{R_s/K_s} - 1)d_s}{\eta_{\text{PA}} a_s^2} + \mu_s |\mathcal{A}_s| \right) > 0, \tag{23}$$

so the unique minimizer is $\beta_s^* = 1$. If $a_s = 0$ and $b_s > 0$ (i.e., side s is fully active), then

$$\Psi''_s(\beta_s) = \frac{6K_s(2^{R_s/K_s} - 1)c_s}{\eta_{\text{PA}} b_s^2 \beta_s^4} + 2\mu_s |\mathcal{A}_s| > 0, \tag{24}$$

hence $\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\text{max}}]$ and has a unique minimizer.

Proposition 2. Assume the mixed case $a_s > 0$, $b_s > 0$, and $d_s > 0$. If

$$\beta_{\text{max}} \leq \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}, \tag{25}$$

then $\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\text{max}}]$. Consequently, the side-wise power-minimizing gain is unique and can be found by checking the two boundaries and the unique root of (21).

Proof: Under (25), the bracketed term in (22) is non-negative over $[1, \beta_{\text{max}}]$. Moreover, it is affine in β_s and therefore can vanish at most at one point. Hence $\Psi''_s(\beta_s)$ is non-negative everywhere and positive on every nontrivial

subinterval of $[1, \beta_{\max}]$, which implies that $\Psi'_s(\beta_s)$ is strictly increasing and $\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$. The minimum of a differentiable strictly convex function on a closed interval is unique and is attained either at an endpoint or at the unique stationary point, namely the unique root of (21). ■

In the numerical study, we use the closed-form SE-optimal rule (17) inside each ϵ -constraint subproblem for the mixed case. For the boundary cases above, we set $\beta_s = 1$ when $|\mathcal{A}_s| = 0$ or $b_s = 0$, and otherwise solve the resulting strictly convex scalar problem on $[1, \beta_{\max}]$. Proposition 2 nevertheless explains why the power minimization remains a benign scalar refinement.

C. Extension of the coefficient rules to TS and SS

Since ES is already covered by (14)–(15), we only state the incremental extension for $p \in \{\text{TS}, \text{SS}\}$. Let $\mathcal{P}_s^{(p)}$ and $\mathcal{A}_s^{(p)}$ denote the passive and active support sets under protocol p . Then

$$\begin{aligned} a_s^{(p)} &= \min_k \sum_{n \in \mathcal{P}_s^{(p)}} u_{s,k,n}, & b_s^{(p)} &= \min_k \sum_{n \in \mathcal{A}_s^{(p)}} u_{s,k,n}, \\ c_s^{(p)} &= \sigma_0^2, & d_s^{(p)} &= \max_k \sigma_a^2 \sum_{n \in \mathcal{A}_s^{(p)}} |h_{s,k,n}|^2. \end{aligned}$$

For TS, $\mathcal{P}_s^{(\text{TS})} = \{1, \dots, N\} \setminus \mathcal{A}_s$ and the effective side target becomes R_s/τ_s , so the protocol dependence enters through the slot-rate inflation. For SS, $\mathcal{P}_s^{(\text{SS})} = \mathcal{S}_s \setminus \mathcal{A}_s$ with disjoint sub-surfaces $\mathcal{S}_t \cup \mathcal{S}_r = \{1, \dots, N\}$, so the same template applies after the aperture shrinkage is absorbed into the support sets. Hence the threshold law (19) remains unchanged after replacing (a_s, b_s, c_s, d_s) by $(a_s^{(p)}, b_s^{(p)}, c_s^{(p)}, d_s^{(p)})$.

V. SPARSE-ACTIVATION PARETO BOUNDARY ALGORITHM

A. Outer sparse active-element selection

The threshold law suggests that only elements with large coherent gain and small active-noise and hardware cost should be activated. To make the ranking rule verifiable, consider a candidate element $n \notin \mathcal{A}_s$ and introduce a relaxed inclusion variable $x_{n,s} \in [0, 1]$. For fixed R_s , fixed β_s , and current aggregates (a_s, b_s, c_s, d_s) , define

$$C_s = \frac{K_s(2^{R_s/K_s} - 1)}{\eta_{\text{PA}}}, \quad \Delta u_{n,s} = \sqrt{\rho_s} |g_n| \min_k |h_{s,k,n}|.$$

The exact increment of d_s depends on the maximizing user after inserting element n ; for ranking we use the representative surrogate

$$\Delta d_{n,s} \approx \rho_s \sigma_a^2 \overline{|h_{s,k,n}|^2}, \quad \overline{|h_{s,k,n}|^2} \triangleq \frac{1}{K_s} \sum_{k \in \mathcal{K}_s} |h_{s,k,n}|^2.$$

Under the same conservative surrogate used in (16)–(20), the n -dependent part of the side cost is approximated by

$$\Psi_s(x_{n,s}) \approx C_s \frac{c_s + \beta_s^2(d_s + x_{n,s} \Delta d_{n,s})}{(a_s + b_s \beta_s + (\beta_s - 1)x_{n,s} \Delta u_{n,s})^2} + \mu_s(|\mathcal{A}_s| + x_{n,s} \Delta u_{n,s} (P_{\text{bias}} + f_u P_{\text{sw},1})).$$

Therefore, the first-order increment at $x_{n,s} = 0$ is

$$\left. \frac{\partial \Psi_s}{\partial x_{n,s}} \right|_{x_{n,s}=0} = -\kappa_{1,s} \Delta u_{n,s} + \kappa_{2,s} \Delta d_{n,s} + P_{\text{bias}} + f_u P_{\text{sw},1} + \mu_s(\beta_s^2 - 1),$$

where

$$\kappa_{1,s} = \frac{2C_s(c_s + d_s \beta_s^2)(\beta_s - 1)}{(a_s + b_s \beta_s)^3}, \quad \kappa_{2,s} = \frac{C_s \beta_s^2}{(a_s + b_s \beta_s)^2}.$$

Hence first-order activation is beneficial whenever

$$\frac{\kappa_{1,s} \Delta u_{n,s}}{\kappa_{2,s} \Delta d_{n,s} + P_{\text{bias}} + f_u P_{\text{sw},1} + \mu_s(\beta_s^2 - 1)} > 1.$$

Absorbing the side-wise factors $\kappa_{1,s}$ and $\kappa_{2,s}$ into normalized weights, and leaving the common gain-dependent term to the subsequent exact power evaluation, yields the practical ranking score

$$\xi_{n,s} = \frac{\omega_s \sqrt{\rho_s} |g_n| \min_k |h_{s,k,n}|}{\lambda_1 \rho_s \sigma_a^2 \overline{|h_{s,k,n}|^2} + \lambda_2 (P_{\text{bias}} + f_u P_{\text{sw},1})}, \quad (26)$$

where ω_s , λ_1 , and λ_2 collect the side-wise constants in the first-order test, and ω_s also reflects the current side demand. In the simulations, $\omega_t = \alpha$, $\omega_r = 1 - \alpha$, and the normalized weights are fixed as $\lambda_1 = \sigma_0^{-2}$ and $\lambda_2 = (P_{\text{max}}^{\text{BS}})^{-1}$. Equation (26) is therefore a first-order ranking surrogate rather than the final cost itself; every selected active set is re-evaluated in the inner step through (9), (20), and (5). Our outer layer ranks all elements by $\max_s \xi_{n,s}$ and assigns each selected element to the side achieving the maximum score. This implements the “open where beneficial” rule in a directly reproducible form.

B. Inner gain and rate-split update

For a given active cardinality $L = |\mathcal{A}|$, we search over a 1-D side rate split $R_t = \alpha \epsilon$ and $R_r = (1 - \alpha) \epsilon$. The active subsets $\mathcal{A}_t$ and $\mathcal{A}_r$ are updated by (26). The side-wise gains are then updated case by case: if $|\mathcal{A}_s| = 0$ or $b_s = 0$, we set $\beta_s = 1$; if $a_s = 0$, we solve the strictly convex scalar problem $\min_{\beta_s \in [1, \beta_{\max}]} \Psi_s(\beta_s)$; otherwise we use (17) or Proposition 2. The required BS powers are computed from (9). The final candidate cost is the total power (5). The minimum over all feasible (L, α) gives one boundary point.

C. Convergence and complexity

Algorithm 1 terminates in finitely many steps because both the active-cardinality set $\mathcal{L}$ and the rate-split grid $\mathcal{G}_\alpha$ are finite. With closed-form gain updates, the dominant per-boundary-point complexity is

$$\mathcal{O}(|\mathcal{L}| |\mathcal{G}_\alpha| (N \log N + N(K_t + K_r))),$$

where the $N \log N$ term comes from ranking the candidate active elements. If the scalar refinement of Proposition 2 is added, only a multiplicative factor proportional to the 1-D search depth appears.

Algorithm 1 Sparse-Activation ϵ -Constraint Solver

Require: Target ϵ , gain cap $\beta_{\max}$, search sets $\mathcal{L}$ and $\mathcal{G}_\alpha$

- 1: Initialize best power as $+\infty$
- 2: **for** $L \in \mathcal{L}$ **do**
- 3: **for** $\alpha \in \mathcal{G}_\alpha$ **do**
- 4: Build $\mathcal{A}_t, \mathcal{A}_r$ using score (26)
- 5: Compute (a_s, b_s, c_s, d_s) for $s \in \{t, r\}$
- 6: **for all** $s \in \{t, r\}$ **do**
- 7: **if** $|\mathcal{A}_s| = 0$ or $b_s = 0$ **then**
- 8: Set $\beta_s \leftarrow 1$
- 9: **else if** $a_s = 0$ **then**
- 10: Find the unique minimizer of $\Psi_s(\beta_s)$ on $[1, \beta_{\max}]$
- 11: **else**
- 12: Update β_s using (17) or Proposition 2
- 13: **end if**
- 14: **end for**
- 15: Invert the target rates through (9)
- 16: **if** $R_\Sigma \geq \epsilon$ and all constraints hold **then**
- 17: Evaluate P_{tot} from (5)
- 18: Keep the candidate with minimum P_{tot}
- 19: **end if**
- 20: **end for**
- 21: **end for**
- 22: **return** best feasible candidate

TABLE I
SIMULATION PARAMETERS

Parameter	Value
STAR-RIS elements N	24
Users (K_t, K_r)	(2, 2)
Fading model	Nakagami- m
Nakagami parameters (m_{BR}, m_{RU})	(1.0, 1.0)
ES coefficients (ρ_t, ρ_r)	(0.55, 0.45)
TS slot fractions (τ_t, τ_r)	(0.55, 0.45)
Noise std. (σ_0, σ_a)	(0.26, 0.05)
Maximum active gain $\beta_{\max}$	5
BS power budget $P_{\max}^{\text{BS}}$	5.0 W
Static controller power P_{ctrl}	0.12 W
Bias power P_{bias}	0.016 W per element
Dynamic base and slope $(P_{\text{sw},0}, P_{\text{sw},1})$	(0.01, 0.006) W
Amplifier coefficient $\mu_t = \mu_r$	0.0025 W
PA efficiency η_{PA}	0.38
Monte Carlo trials	100

VI. NUMERICAL RESULTS

We consider a hybrid STAR-RIS with $N = 24$ elements, $K_t = K_r = 2$, Nakagami- m fading, and the normalized power parameters listed in Table I. All curves are averaged over 100 independent channel draws. All figures are generated by the same simulation pipeline. Unless stated otherwise, the BS transmit-power budget is 5.0 W, the ES coefficients are $\rho_t = 0.55$ and $\rho_r = 0.45$, the TS slot fractions are $\tau_t = 0.55$ and $\tau_r = 0.45$, and the gain cap is $\beta_{\max} = 5$.

Fig. 2 shows the EE-SE Pareto boundary. At very low SE, the passive STAR-RIS remains competitive because no active hardware is needed, and at 2 bit/s/Hz it is essentially coincident with the proposed sparse-hybrid design. As the target SE increases, however, the passive benchmark suffers

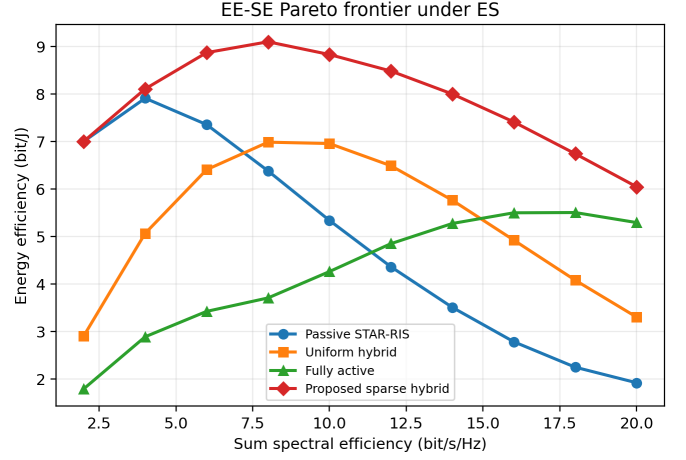


Fig. 2. Average EE-SE Pareto boundary under ES. Sparse activation dominates the medium-to-high-SE region, while the passive design is only competitive at low SE.

a rapidly growing BS-power burden, while the fully active benchmark pays a large amplification cost. The proposed sparse-hybrid design dominates the medium-to-high-SE region and stays feasible up to 20 bit/s/Hz under the considered budget. At 12 bit/s/Hz, the proposed design improves EE by approximately 30.7% over uniform hybrid activation, 74.9% over the fully active design, and 94.4% over the passive design. This supports the practical message of the paper: only a minority of carefully chosen active elements is needed to unlock most of the SE gain.

Fig. 3 compares the cross-mode design implications of the proposed sparse-hybrid solver. TS requires the densest active set because the effective slot targets are inflated by $1/\tau_s$; at 12 bit/s/Hz its active ratio is about 0.54, versus about 0.27 for ES and 0.29 for SS. In terms of activation density, SS tracks ES much more closely than TS over the whole range while avoiding the severe slot-rate penalty. The gain curves reveal a related but not identical regime split: SS stays at or near the gain cap over most of the tested range, TS also uses near-cap gains but with much denser activation, whereas ES ramps its side-wise gains more gradually before saturating near the high-SE end.

Fig. 5 compares the power decomposition at the representative target $R_\Sigma = 12$ bit/s/Hz under ES. The fully active design indeed minimizes the BS transmit-power share, about 0.38 W on average, but it pays excessive static/dynamic and amplification costs, with the amplification term alone reaching about 1.44 W. The passive design has no amplification cost but suffers from a large BS-power term of about 2.78 W due to the double fading. The proposed sparse-hybrid scheme achieves the lowest total power, about 1.44 W versus about 1.91 W for uniform hybrid, by sharply reducing the BS radiated-power requirement while keeping the amplification term well below the fully active benchmark. This decomposition clarifies why uniform or all-active operation is not EE-optimal.

Fig. 4 compares the hardware sensitivity across ES, TS, and SS at the representative target $R_\Sigma = 12$ bit/s/Hz. In Fig. 4(a), the average side-wise gain decreases with σ_a for

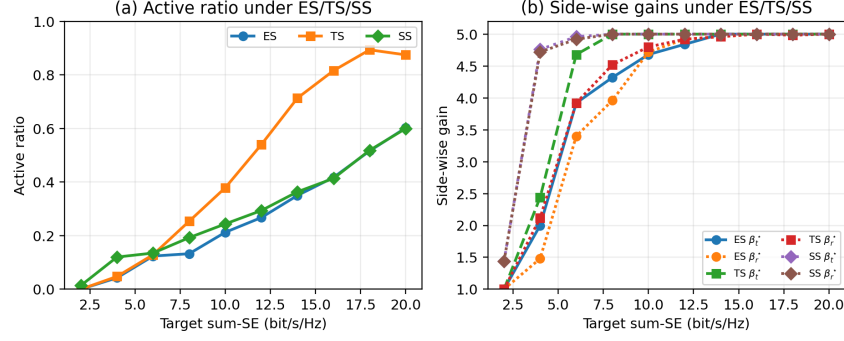


Fig. 3. Design implications of the proposed sparse-hybrid solver across ES, TS, and SS: (a) active ratio versus target SE; (b) side-wise common gains.

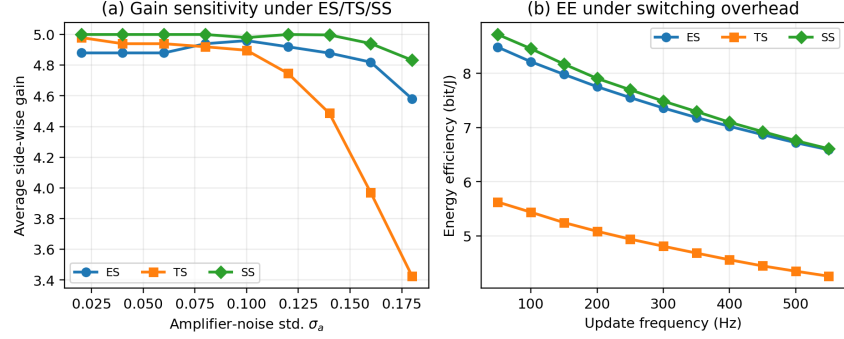


Fig. 4. Hardware sensitivity across ES, TS, and SS: (a) average side-wise gain versus amplifier-noise level; (b) EE under switching overhead. TS becomes fragile under strong amplifier noise, while ES and SS remain feasible over the tested range.

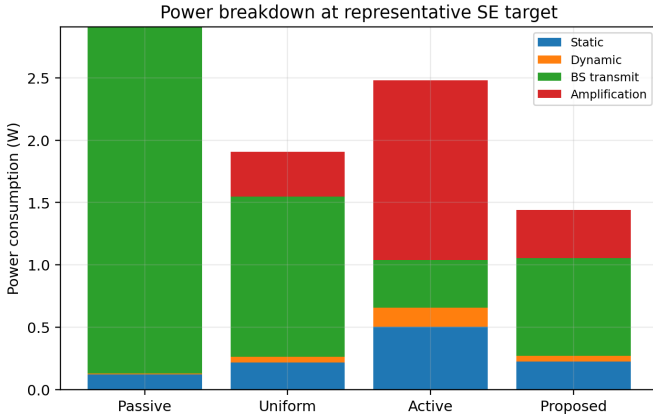


Fig. 5. Average power breakdown at $R_\Sigma = 12$ bit/s/Hz under ES. Sparse activation balances BS transmit and amplification power more efficiently than the benchmarks.

all three protocols, with the strongest drop appearing under TS because the slot-rate inflation already forces a denser active set. SS remains closest to the gain cap over the tested range, ES degrades more moderately, and TS backs off its preferred gain much earlier. Fig. 4(b) shows that increasing the state-update frequency from 50 Hz to 550 Hz reduces EE for all three protocols; SS stays slightly above ES throughout the sweep, and TS stays consistently lower because its denser activation pattern magnifies the dynamic-cost penalty.

Two additional remarks are worth emphasizing. First, the

asymmetry between the transmission and reflection sides mainly enters through (a_s, b_s, d_s) and through the rate split α ; hence the present framework captures side-wise asymmetric rate demands. Extending the analysis to heterogeneous user-specific QoS or explicit worst-user constraints would require reformulating the power-inversion step in (9) and the side-cost model in (20), and is left for future work. Second, the same active elements can be reused for single-RF calibration or channel sounding: one can augment the score in (26) with an estimation utility term and impose $|\mathcal{A}| \geq L_{\text{cal}}$, which provides a direct path toward joint calibration and forwarding hardware.

VII. CONCLUSION

This paper presented a structured study of the EE-SE tradeoff in hybrid active-passive STAR-RIS. The key contribution is not merely another numerical solver, but an interpretable Pareto-boundary description connecting amplifier noise, active-set sparsity, side-wise task asymmetry, and hardware power consumption. The analysis yielded a closed-form SE-optimal gain, an active-on threshold, and a simple sparse-activation rule, while the ϵ -constraint algorithm translated these results into a reproducible Pareto-boundary generator. The numerical results showed that the best operating point is rarely fully passive or fully active: low-SE regimes favor passive STAR-RIS, whereas medium-to-high-SE regimes favor a sparse active subset with carefully controlled gains. Across the studied STAR protocols, TS requires denser activation and is more sensitive to amplifier noise, whereas ES and SS exhibit more gradual gain adaptation. Future work includes

exact robust designs with imperfect CSI, joint calibration and forwarding architectures with a single RF chain, and learning-assisted online control for rapidly time-varying STAR-RIS states.

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